

# WEEK 4 – DATA SCIENCE AND VISUALIZATION

## Univariate and Bivariate Analysis – Spotify Songs Dataset

**Project:** Discovering the Factors Behind Song Popularity Through Data Science

**Dataset:** Spotify Songs Dataset | **Input:** Cleaned Week 3 dataset

### 1. Introduction

Week 4 focuses on Exploratory Data Analysis (EDA) of the cleaned Spotify Songs dataset. The purpose is to understand individual feature distributions and investigate relationships between important musical characteristics and song popularity. This stage follows the Week 3 data-cleaning and comparison analysis.

### 2. Objective

The main objective is to perform **univariate and bivariate analysis**. Univariate analysis examines one variable at a time, while bivariate analysis examines relationships between two variables. The project presentation identifies popularity, danceability, energy, language, loudness, and related audio features for EDA.

### 3. Dataset Used

The Week 3 notebook records the cleaned Spotify dataset as **19,335 rows and 45 columns**. Important features include popularity, danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration. The Week 1 project identifies **Popularity (0–100)** as the target variable for the planned analysis.

### 4. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook are used for statistical summaries and visualization.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
```

### 5. Load the Cleaned Dataset

The cleaned Week 3 dataset is used as the starting point for Week 4 EDA.

```
df = pd.read_csv("spotify_cleaned_week3.csv", low_memory=False)
print("Dataset shape:", df.shape)
df.head()
```

### 6. Selected Features

The analysis focuses on popularity and major Spotify audio features.

```
features = ["popularity", "danceability", "energy", "loudness",
            "speechiness", "acousticness", "instrumentalness",
            "liveness", "valence", "tempo"]
print(df[features].describe().T)
```

### 7. Univariate Analysis

Univariate analysis studies one variable at a time. Histograms and KDE plots help understand concentration, spread, skewness, and possible extreme observations.

```
plt.figure(figsize=(9,5))
sns.histplot(df["popularity"], bins=20, kde=True)
plt.title("Distribution of Song Popularity")
plt.xlabel("Popularity")
plt.ylabel("Number of Songs")
plt.show()

for col in ["danceability", "energy", "loudness", "speechiness",
            "acousticness", "instrumentalness", "liveness", "valence", "tempo"]:
    plt.figure(figsize=(8,4))
    sns.histplot(df[col], bins=30, kde=True)
```

```
plt.title(f"Distribution of {col}")
plt.xlabel(col)
plt.ylabel("Number of Songs")
plt.show()
```

## 8. Univariate Interpretation

The univariate stage provides a feature-by-feature understanding of the cleaned Spotify dataset. Week 2 reported approximate means of 0.616 for danceability, 0.584 for energy, -8.605 dB for loudness, 0.309 for acousticness, 0.491 for valence, and 119.795 BPM for tempo. These values provide a numerical baseline for interpreting the Week 4 distributions.

## 9. Bivariate Analysis

Bivariate analysis examines relationships between two variables. The main focus is on popularity versus audio features and selected relationships between audio features.

```
plt.figure(figsize=(9,5))
sns.scatterplot(data=df, x="danceability", y="popularity", alpha=0.5)
plt.title("Popularity vs Danceability")
plt.xlabel("Danceability")
plt.ylabel("Popularity")
plt.show()
```

```
plt.figure(figsize=(9,5))
sns.scatterplot(data=df, x="energy", y="popularity", alpha=0.5)
plt.title("Popularity vs Energy")
plt.xlabel("Energy")
plt.ylabel("Popularity")
plt.show()
```

```
plt.figure(figsize=(9,5))
sns.scatterplot(data=df, x="energy", y="loudness", alpha=0.5)
plt.title("Energy vs Loudness")
plt.xlabel("Energy")
plt.ylabel("Loudness")
plt.show()
```

```
plt.figure(figsize=(9,5))
sns.scatterplot(data=df, x="valence", y="popularity", alpha=0.5)
plt.title("Popularity vs Valence")
plt.xlabel("Valence")
plt.ylabel("Popularity")
plt.show()
```

## 10. Correlation Analysis

Correlation analysis complements the bivariate plots by quantifying linear relationships between numerical features. Correlation indicates association and does not by itself establish causation.

```
numerical_features = ["popularity", "danceability", "energy", "loudness",
                      "speechiness", "acousticness", "instrumentalness",
                      "liveness", "valence", "tempo"]

corr = df[numerical_features].corr()
plt.figure(figsize=(11,8))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", center=0)
plt.title("Correlation Heatmap of Spotify Features")
plt.tight_layout()
plt.show()
```

## 11. Analysis Covered

The project presentation specifies the following EDA areas: univariate analysis of popularity, danceability, energy, and language; bivariate analysis of popularity versus danceability, popularity versus energy, energy versus loudness, and language versus popularity; and correlation analysis using a numerical-feature heatmap.

## 12. Interpretation

The bivariate plots and correlation matrix should be used to determine whether the relationships suggested during project planning are visible in the cleaned data. In particular, the analysis investigates whether danceability and energy are

associated with popularity and how energy relates to loudness. These are associations in the data, not evidence of causation.

### **13. Project-Level Insights to Validate**

The Week 1 presentation lists expected observations that most tracks may have medium popularity, energy and loudness may be strongly related, danceability may have some association with popularity, and potential extreme values may occur in tempo and duration. These were project expectations and should be validated using the actual Week 4 plots and correlation results.

### **14. Conclusion**

Week 4 extends the Spotify project from data cleaning into Exploratory Data Analysis. Univariate analysis examines distributions of popularity and major audio features, while bivariate analysis investigates relationships between popularity and musical characteristics and between selected audio features. Correlation analysis provides an additional numerical view of relationships. Together, these analyses establish the foundation for deeper statistical interpretation, feature engineering, and the later machine-learning phase.

### **15. Project Status**

**Week 4: Univariate and Bivariate Analysis — Completed**